Magnet vs Traditional Schools: Full Analysis

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Introduction

We model:

$$P(Y = 1|X) = \frac{e^{\eta}}{1 + e^{\eta}}, \quad \eta = \beta_0 + \beta_1 X_1 + \cdots$$

Load Data

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)

mt <- read.csv("MagTrad.csv")

mt <- mt %>%
  mutate(
    DEMOGRAPHIC = trimws(DEMOGRAPHIC),
    DEMOGRAPHIC_CATEGORY = trimws(DEMOGRAPHIC_CATEGORY)
  )
```

Create Variables

```
mt <- mt %>%
  mutate(
    race_df = case_when(
      str_detect(DEMOGRAPHIC, regex("black", TRUE)) ~ "Black",
      str_detect(DEMOGRAPHIC, regex("white", TRUE)) ~ "White",
```

```r
      str_detect(DEMOGRAPHIC, regex("asian", TRUE)) ~ "Asian",
      str_detect(DEMOGRAPHIC, regex("hispanic", TRUE)) ~ "Hispanic",
      str_detect(DEMOGRAPHIC, regex("multi", TRUE)) ~ "Multiracial",
      str_detect(DEMOGRAPHIC, regex("indian", TRUE)) ~ "Indigenous American",
      TRUE ~ "Other"
    )
  )

mt$race_df <- relevel(as.factor(mt$race_df), ref="White")

mt <- mt %>%
  mutate(
    ses_clean = case_when(
      str_detect(DEMOGRAPHIC, regex("non free", TRUE)) ~ "Non-Free",
      str_detect(DEMOGRAPHIC, regex("free", TRUE)) ~ "Free",
      TRUE ~ NA_character_
    )
  )
```

EDA

Race Summary

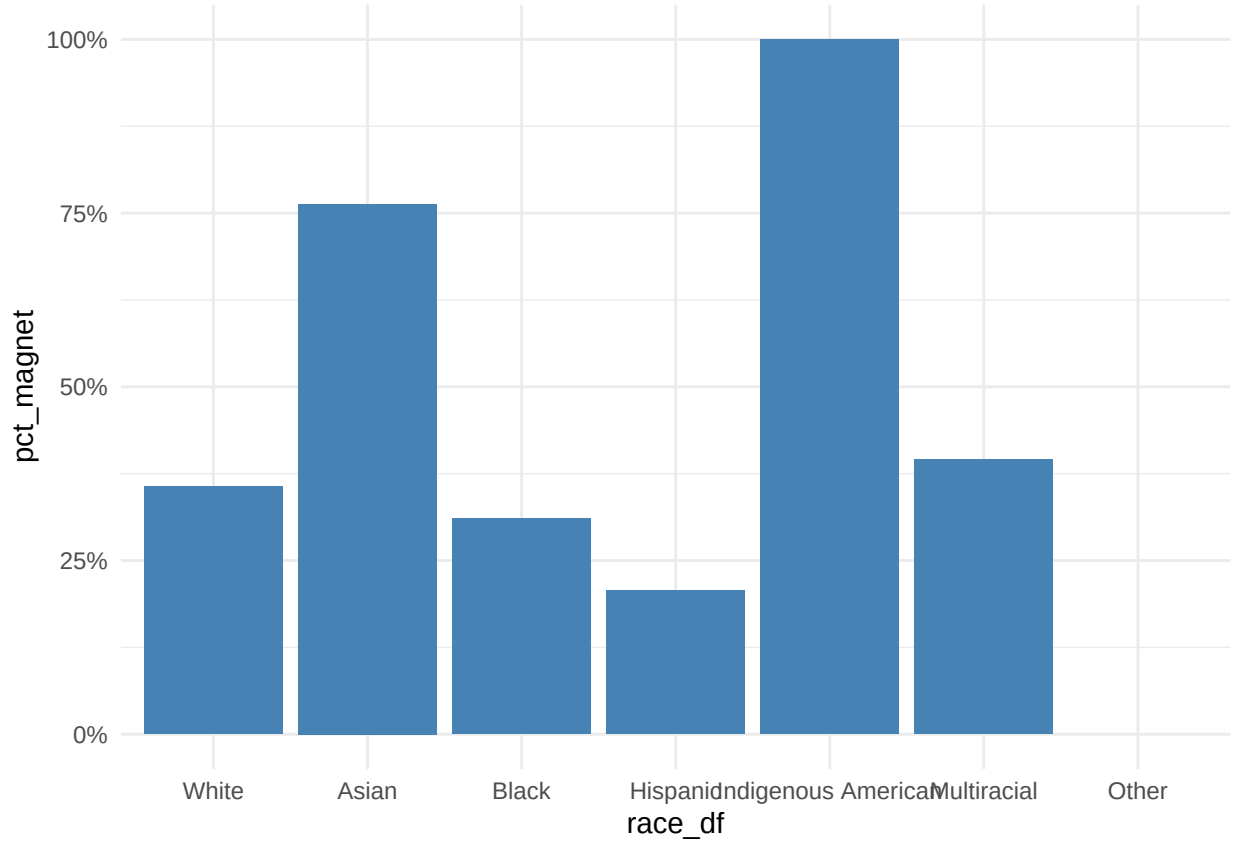
```r
race_summary <- mt %>%
  filter(DEMOGRAPHIC_CATEGORY=="Ethnicity") %>%
  group_by(race_df) %>%
  summarise(
    magnet=sum(MAGNET_STUDENTS),
    traditional=sum(TRADITIONAL_STUDENTS),
    total=magnet+traditional,
    pct_magnet=magnet/total
  )

race_summary
```

```
## # A tibble: 7 x 5
##   race_df             magnet traditional total pct_magnet
##   <fct>                <dbl>       <dbl> <dbl>      <dbl>
## 1 White                11073       19985 31058      0.357
## 2 Asian                 6832        2120  8952      0.763
## 3 Black                13104       29137 42241      0.310
## 4 Hispanic             13345       51232 64577      0.207
## 5 Indigenous American     10           0    10      1
## 6 Multiracial           2678        4099  6777      0.395
## 7 Other                    0           0     0    NaN
```

```r
ggplot(race_summary, aes(x=race_df, y=pct_magnet)) +
  geom_col(fill="steelblue") +
  scale_y_continuous(labels=scales::percent) +
  theme_minimal()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
```

```
## (`geom_col()`).
```



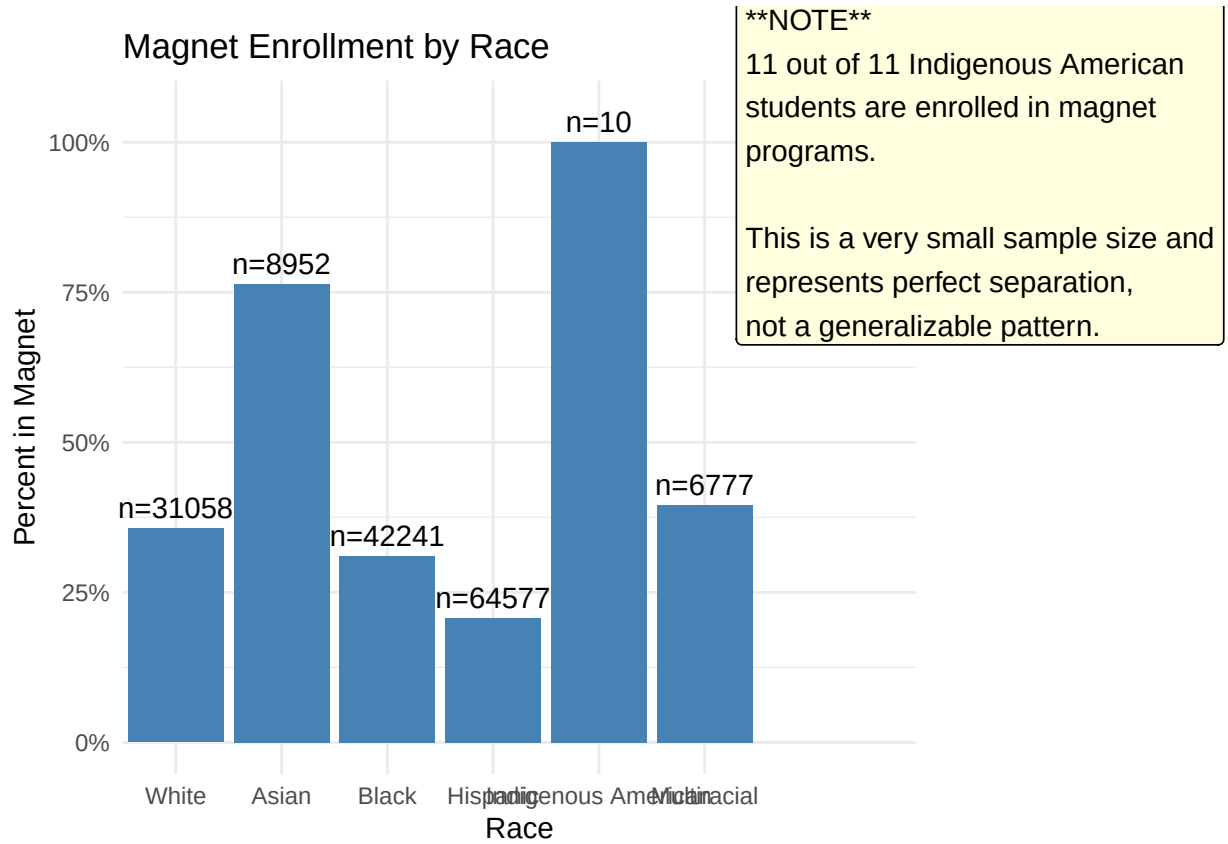
Indigenous Americans represent 11 Magnet students and 0 traditional students in this data set. Below we remove this group to visualize the groups that have more visual interpretitability

```r
library(ggplot2)
library(dplyr)
library(scales)
```

```
##
## Attaching package: 'scales'

## The following object is masked from 'package:purrr':
##
##     discard

## The following object is masked from 'package:readr':
##
##     col_factor
```

```r
# Clean data (remove invalid rows like "Other")
race_summary_clean <- race_summary %>%
  filter(!is.na(pct_magnet), total > 0)

# Create factor order (optional but cleaner)
```

```r
race_summary_clean$race_df <- factor(
  race_summary_clean$race_df,
  levels = c("White", "Asian", "Black", "Hispanic", "Indigenous American", "Multiracial")
)

ggplot(race_summary_clean, aes(x = race_df, y = pct_magnet)) +

  # Bars
  geom_col(fill = "steelblue") +

  # Sample size labels
  geom_text(aes(label = paste0("n=", total)),
            vjust = -0.5,
            size = 4) +

  # Expand x-axis to create space for annotation
  scale_x_discrete(expand = expansion(add = c(0.5, 2))) +

  # Percent axis
  scale_y_continuous(labels = percent_format(), limits = c(0,1.05)) +

  # Annotation box OUTSIDE bars
  annotate(
    "label",
    x = 6.3, y = 0.95,
    label = "**NOTE**\n11 out of 11 Indigenous American\nstudents are enrolled in magnet\nprograms.\n\n
    hjust = 0,
    size = 4,
    fill = "lightyellow",
    color = "black",
    linewidth = 0.3
  ) +

  labs(
    title = "Magnet Enrollment by Race",
    x = "Race",
    y = "Percent in Magnet"
  ) +

  theme_minimal() +

  # Prevent clipping of annotation
  coord_cartesian(clip = "off") +

  # Add margin so box is visible
  theme(
    plot.margin = margin(10, 120, 10, 10)
  )
```

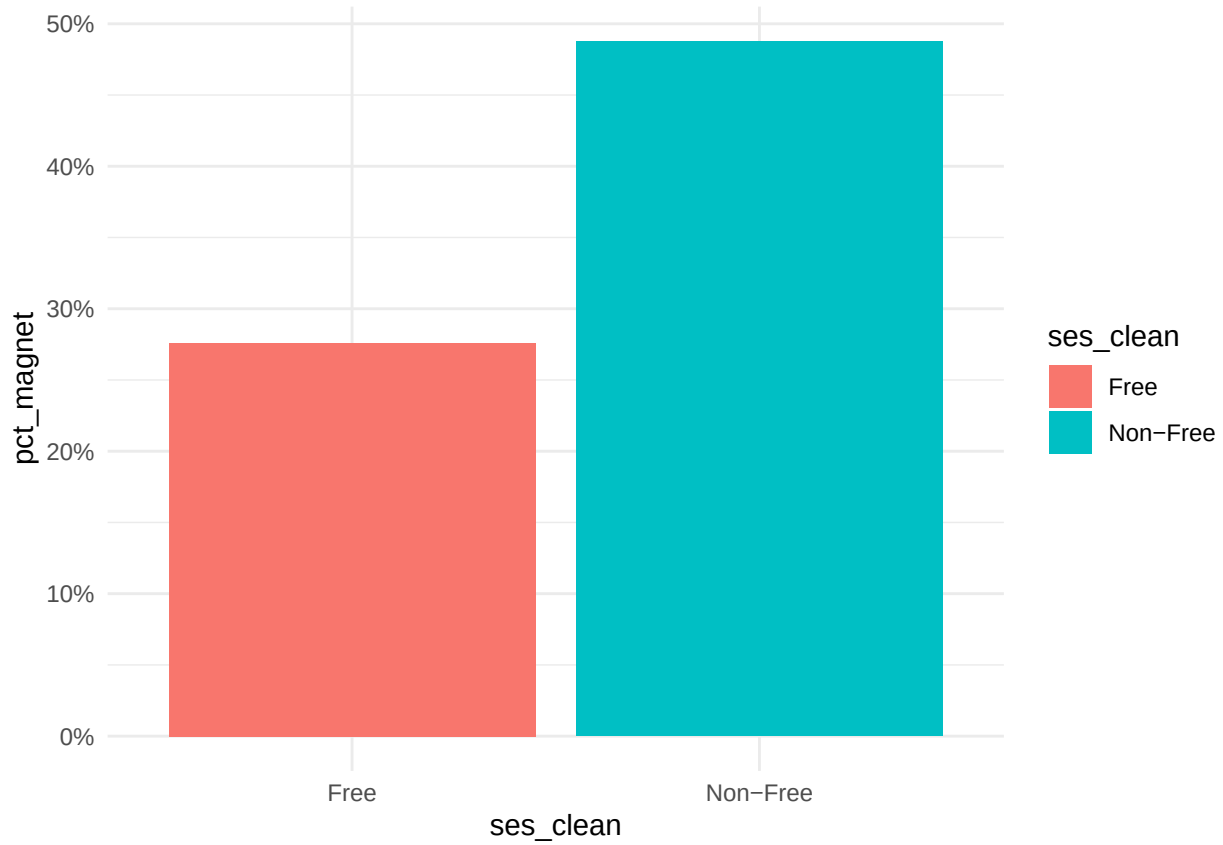


SES Summary

```
ses_summary <- mt %>%
  filter(DEMOGRAPHIC_CATEGORY=="Meal Status") %>%
  group_by(ses_clean) %>%
  summarise(
    magnet=sum(MAGNET_STUDENTS),
    traditional=sum(TRADITIONAL_STUDENTS),
    total=magnet+traditional,
    pct_magnet=magnet/total
  ) %>%
  filter(!is.na(ses_clean))

ses_summary
```

```
## # A tibble: 2 x 5
##   ses_clean magnet traditional total pct_magnet
##   <chr>      <dbl>       <dbl> <dbl>      <dbl>
## 1 Free       23325       61203 84528      0.276
## 2 Non-Free    7974        8377 16351      0.488
```

```
ggplot(ses_summary, aes(x=ses_clean, y=pct_magnet, fill=ses_clean)) +
  geom_col() +
  scale_y_continuous(labels=scales::percent) +
  theme_minimal()
```



School-Level Data

```
race_school <- mt %>%
  filter(DEMOGRAPHIC_CATEGORY=="Ethnicity") %>%
  group_by(SCHOOL_NAME,SCHOOL_YEAR,race_df) %>%
  summarise(
    magnet=sum(MAGNET_STUDENTS),
    traditional=sum(TRADITIONAL_STUDENTS),
    .groups="drop"
  ) %>%
  filter((magnet+traditional)>0) %>%
  filter(race_df!="Indigenous American")
```

Train/Test Split

```
set.seed(42)
train_idx <- sample(1:nrow(race_school),0.7*nrow(race_school))

train <- race_school[train_idx,]
test <- race_school[-train_idx,]
```

Model 1: Race Only

```r
log.obj <- glm(
  cbind(magnet,traditional)~race_df,
  family=binomial,
  data=train
)

summary(log.obj)
```

```
##
## Call:
## glm(formula = cbind(magnet, traditional) ~ race_df, family = binomial,
##     data = train)
##
## Coefficients:
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)         -0.73057    0.01415 -51.613   <2e-16 ***
## race_dfAsian         1.84916    0.03368  54.911   <2e-16 ***
## race_dfBlack         0.18876    0.01851  10.200   <2e-16 ***
## race_dfHispanic     -0.66290    0.01885 -35.165   <2e-16 ***
## race_dfMultiracial   0.29820    0.03281   9.088   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 47360  on 328  degrees of freedom
## Residual deviance: 39403  on 324  degrees of freedom
## AIC: 40861
##
## Number of Fisher Scoring iterations: 5
```

```r
exp(coef(log.obj))
```

```
##        (Intercept)       race_dfAsian       race_dfBlack    race_dfHispanic
##          0.4816335          6.3544605          1.2077558          0.5153552
## race_dfMultiracial
##          1.3474262
```

Interpretation (All Coefficients)

Intercept (White)

$$\hat{\beta}_0 = -0.73057$$

$$e^{-0.73057} = 0.4816$$

$$p = \frac{0.4816}{1 + 0.4816} = 0.325$$

White students have a 32.5 percent probability of magnet enrollment.

Asian

$$e^{1.84916} = 6.354$$

$$6.354 - 1 = 5.354$$

$$5.354 \times 100 = 535.4\%$$

Asian students have 535.4 percent higher odds than White students.

Black

$$e^{0.18876} = 1.2078$$

$$1.2078 - 1 = 0.2078$$

$$0.2078 \times 100 = 20.78\%$$

Black students have 20.78 percent higher odds than White students.

Hispanic

$$e^{-0.66290} = 0.515$$

$$1 - 0.515 = 0.485$$

$$0.485 \times 100 = 48.5\%$$

Hispanic students have 48.5 percent lower odds.

Multiracial

$$e^{0.29820} = 1.347$$

$$1.347 - 1 = 0.347$$

$$0.347 \times 100 = 34.7\%$$

Multiracial students have 34.7 percent higher odds.

Model 2: Race + SES

```
school_ses <- mt %>%
  filter(DEMOGRAPHIC_CATEGORY=="Meal Status") %>%
  group_by(SCHOOL_NAME,SCHOOL_YEAR,ses_clean) %>%
  summarise(total=sum(MAGNET_STUDENTS+TRADITIONAL_STUDENTS),.groups="drop") %>%
  filter(!is.na(ses_clean)) %>%
  group_by(SCHOOL_NAME,SCHOOL_YEAR) %>%
  slice_max(total,n=1) %>%
  ungroup()

race_ses <- race_school %>%
  left_join(school_ses,by=c("SCHOOL_NAME","SCHOOL_YEAR"))

log.obj.2 <- glm(
  cbind(magnet,traditional)~race_df+ses_clean,
  family=binomial,
  data=race_ses
)

summary(log.obj.2)
```

```
##
## Call:
## glm(formula = cbind(magnet, traditional) ~ race_df + ses_clean,
##     family = binomial, data = race_ses)
##
## Coefficients:
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)         -0.59175    0.01470 -40.258  < 2e-16 ***
## race_dfAsian         1.73700    0.03363  51.650  < 2e-16 ***
## race_dfBlack        -0.22351    0.01957 -11.423  < 2e-16 ***
## race_dfHispanic     -0.72426    0.01890 -38.313  < 2e-16 ***
## race_dfMultiracial   0.11777    0.03322   3.546 0.000392 ***
## ses_cleanNon-Free    0.06123    0.03515   1.742 0.081516 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```
##     Null deviance: 42946  on 305  degrees of freedom
## Residual deviance: 35553  on 300  degrees of freedom
##   (165 observations deleted due to missingness)
## AIC: 36938
##
## Number of Fisher Scoring iterations: 5
```

```r
exp(coef(log.obj.2))
```

```
##        (Intercept)       race_dfAsian       race_dfBlack    race_dfHispanic
##          0.5533583          5.6802591          0.7997090          0.4846816
## race_dfMultiracial  ses_cleanNon-Free
##          1.1249889          1.0631409
```

SES Interpretation

$$e^{0.06123} = 1.063$$

$$1.063 - 1 = 0.063$$

$$0.063 \times 100 = 6.3\%$$

Non-Free students have 6.3 percent higher odds, not statistically significant.

Model 3: Interaction

```r
log.obj.int <- glm(
  cbind(magnet,traditional)~race_df*ses_clean,
  family=binomial,
  data=race_ses
)

summary(log.obj.int)
```

```
##
## Call:
## glm(formula = cbind(magnet, traditional) ~ race_df * ses_clean,
##     family = binomial, data = race_ses)
##
## Coefficients:
##                                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)                        -0.62405    0.01512 -41.274  < 2e-16 ***
## race_dfAsian                        1.80985    0.03512  51.529  < 2e-16 ***
## race_dfBlack                       -0.16945    0.02005  -8.452  < 2e-16 ***
## race_dfHispanic                    -0.69909    0.01943 -35.975  < 2e-16 ***
## race_dfMultiracial                  0.17358    0.03463   5.013 5.37e-07 ***
## ses_cleanNon-Free                   0.39974    0.04830   8.276  < 2e-16 ***
## race_dfAsian:ses_cleanNon-Free     -0.92465    0.12399  -7.458 8.81e-14 ***
```

```
## race_dfBlack:ses_cleanNon-Free       -1.78802    0.14729 -12.140  < 2e-16 ***
## race_dfHispanic:ses_cleanNon-Free    -0.01468    0.09280  -0.158    0.874
## race_dfMultiracial:ses_cleanNon-Free -0.64241    0.12410  -5.177 2.26e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 42946  on 305  degrees of freedom
## Residual deviance: 35301  on 296  degrees of freedom
##   (165 observations deleted due to missingness)
## AIC: 36693
##
## Number of Fisher Scoring iterations: 5
```

```r
exp(coef(log.obj.int))
```

```
##                          (Intercept)                        race_dfAsian
##                            0.5357684                           6.1095188
##                          race_dfBlack                      race_dfHispanic
##                            0.8441313                           0.4970372
##                   race_dfMultiracial                    ses_cleanNon-Free
##                            1.1895537                           1.4914381
##       race_dfAsian:ses_cleanNon-Free       race_dfBlack:ses_cleanNon-Free
##                            0.3966705                           0.1672910
##    race_dfHispanic:ses_cleanNon-Free race_dfMultiracial:ses_cleanNon-Free
##                            0.9854318                           0.5260217
```

Interaction Interpretation

$$\eta = \beta_0 + \beta_1 + \beta_2 + \beta_3$$

Combined effects must be interpreted jointly.

ANOVA

```r
anova(log.obj.2, log.obj.int, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: cbind(magnet, traditional) ~ race_df + ses_clean
## Model 2: cbind(magnet, traditional) ~ race_df * ses_clean
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1       300      35553
## 2       296      35301  4   252.32 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

Reject H0 if p-value < 0.05. Interaction improves model.

Validation

```
test$pred <- predict(log.obj,newdata=test,type="response")
test$actual <- test$magnet/(test$magnet+test$traditional)

mean(abs(test$pred-test$actual))
```

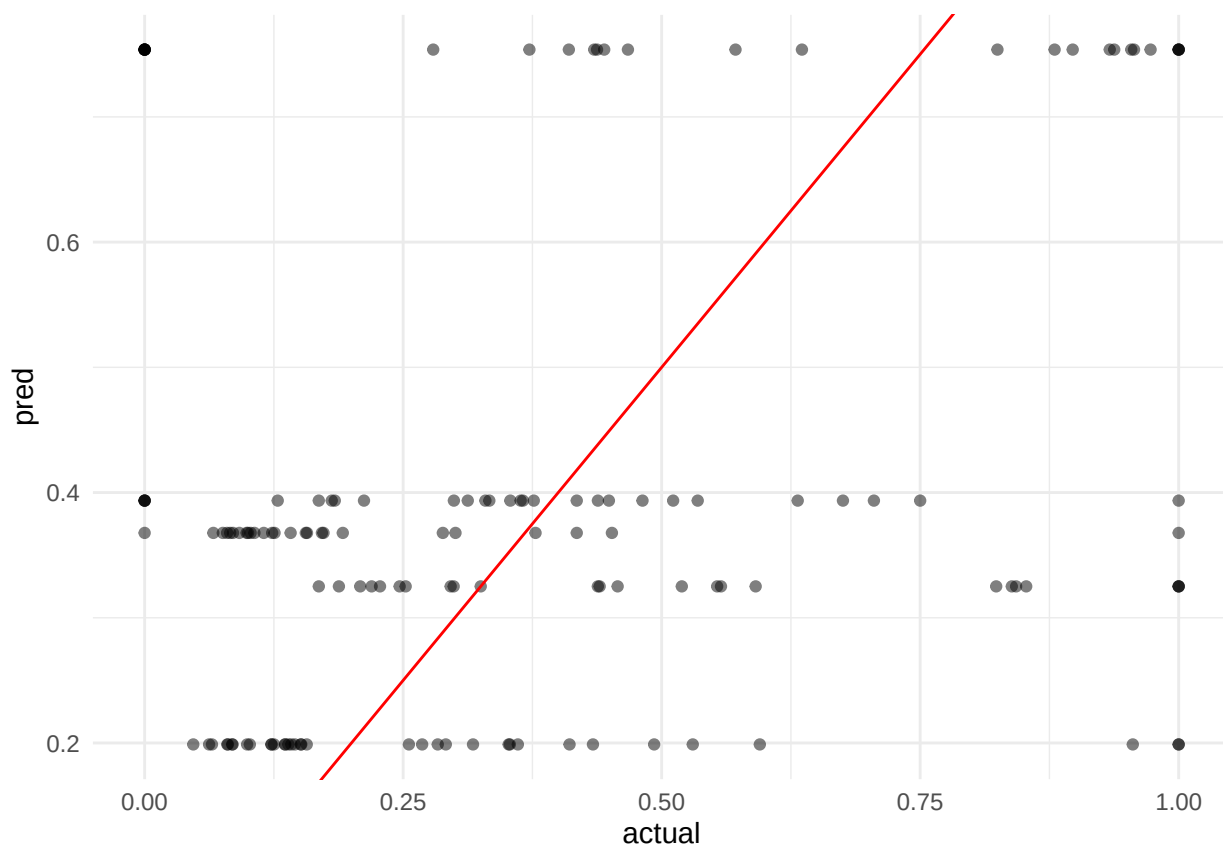
```
## [1] 0.2418969
```

```
sqrt(mean((test$pred-test$actual)^2))
```

```
## [1] 0.3158964
```

Visualization: Predicted vs Actual

```
ggplot(test, aes(x=actual, y=pred)) +
  geom_point(alpha=0.5) +
  geom_abline(slope=1, intercept=0, color="red") +
  theme_minimal()
```



build magnet programs variable

```r
# ================================
# 1. BUILD RACE SCHOOL DATASET
# ================================

race_school_pct <- mt %>%
  filter(DEMOGRAPHIC_CATEGORY == "Ethnicity") %>%

  group_by(SCHOOL_NAME, SCHOOL_YEAR, race_df) %>%
  summarise(
    magnet = sum(MAGNET_STUDENTS),
    traditional = sum(TRADITIONAL_STUDENTS),
    total_race = magnet + traditional,
    .groups = "drop"
  ) %>%

  group_by(SCHOOL_NAME, SCHOOL_YEAR) %>%
  mutate(
    total_school = sum(total_race),
    pct_race = total_race / total_school,
    pct_magnet = magnet / total_race,
    pct_traditional = traditional / total_race
  ) %>%

  ungroup()

race_school_pct
```

```
## # A tibble: 606 x 10
##    SCHOOL_NAME    SCHOOL_YEAR race_df magnet traditional total_race total_school
##    <chr>                <dbl> <fct>    <dbl>       <dbl>      <dbl>        <dbl>
##  1 ALONSO HIGH S~        2020 White       46         668        714         2413
##  2 ALONSO HIGH S~        2020 Asian       16         100        116         2413
##  3 ALONSO HIGH S~        2020 Black        0         157        157         2413
##  4 ALONSO HIGH S~        2020 Hispan~     52        1273       1325         2413
##  5 ALONSO HIGH S~        2020 Indige~      0           0          0         2413
##  6 ALONSO HIGH S~        2020 Multir~      0         101        101         2413
##  7 ALONSO HIGH S~        2021 White       72         638        710         2516
##  8 ALONSO HIGH S~        2021 Asian       36          93        129         2516
##  9 ALONSO HIGH S~        2021 Black        0         159        159         2516
## 10 ALONSO HIGH S~        2021 Hispan~     92        1321       1413         2516
## # i 596 more rows
## # i 3 more variables: pct_race <dbl>, pct_magnet <dbl>, pct_traditional <dbl>
```

```r
# 2. ADD DOMINANT RACE
dominant_race <- race_school_pct %>%
  group_by(SCHOOL_NAME, SCHOOL_YEAR) %>%
  slice_max(pct_race, n = 1) %>%
  ungroup() %>%
  select(SCHOOL_NAME, SCHOOL_YEAR, dominant_race = race_df)
dominant_race
```

```
## # A tibble: 99 x 3
##    SCHOOL_NAME           SCHOOL_YEAR dominant_race
```

```
##    <chr>                        <dbl> <fct>
##  1 "ALONSO HIGH SCHOOL"          2020 Hispanic
##  2 "ALONSO HIGH SCHOOL"          2021 Hispanic
##  3 "ALONSO HIGH SCHOOL"          2022 Hispanic
##  4 "ALONSO HIGH SCHOOL"          2023 Hispanic
##  5 "ARMWOOD HIGH SCHOOL"         2020 Black
##  6 "ARMWOOD HIGH SCHOOL"         2021 Black
##  7 "ARMWOOD HIGH SCHOOL"         2022 Black
##  8 "ARMWOOD HIGH SCHOOL"         2023 Black
##  9 "Alonso High School"          2024 Hispanic
## 10 "Alonso High School "         2025 Hispanic
## # i 89 more rows
```

```
race_school_pct <- race_school_pct %>%
  left_join(dominant_race, by = c("SCHOOL_NAME", "SCHOOL_YEAR"))

race_school_pct
```

```
## # A tibble: 606 x 11
##    SCHOOL_NAME    SCHOOL_YEAR race_df magnet traditional total_race total_school
##    <chr>                <dbl> <fct>    <dbl>       <dbl>      <dbl>        <dbl>
##  1 ALONSO HIGH S~        2020 White       46         668        714         2413
##  2 ALONSO HIGH S~        2020 Asian       16         100        116         2413
##  3 ALONSO HIGH S~        2020 Black        0         157        157         2413
##  4 ALONSO HIGH S~        2020 Hispan~     52        1273       1325         2413
##  5 ALONSO HIGH S~        2020 Indige~      0           0          0         2413
##  6 ALONSO HIGH S~        2020 Multir~      0         101        101         2413
##  7 ALONSO HIGH S~        2021 White       72         638        710         2516
##  8 ALONSO HIGH S~        2021 Asian       36          93        129         2516
##  9 ALONSO HIGH S~        2021 Black        0         159        159         2516
## 10 ALONSO HIGH S~        2021 Hispan~     92        1321       1413         2516
## # i 596 more rows
## # i 4 more variables: pct_race <dbl>, pct_magnet <dbl>, pct_traditional <dbl>,
## #   dominant_race <fct>
```

```
# 3. MAGNET PROGRAM DATASET
magnet_programs <- data.frame(
  SCHOOL_NAME = c(
    "BLAKE HIGH SCHOOL","KING HIGH SCHOOL","MIDDLETON HIGH SCHOOL",
    "TAMPA BAY TECH HIGH SCHOOL","STRAWBERRY CREST HIGH SCHOOL",
    "ALONSO HIGH SCHOOL","ROBINSON HIGH SCHOOL","PLANT HIGH SCHOOL",
    "SPOTO HIGH SCHOOL","WHARTON HIGH SCHOOL",
    "STEWART MIDDLE MAGNET SCHOOL",
    "PROGRESS VILLAGE MIDDLE MAGNET SCHOOL",
    "DOWDELL MIDDLE MAGNET SCHOOL"
  ),
  STEM = c(0,0,1,1,0,0,0,0,0,0,1,0,1),
  ARTS = c(1,0,0,0,0,0,0,0,0,0,0,1,0),
  IB = c(0,1,0,0,1,1,1,1,0,0,0,0,0),
  MEDICAL = c(0,0,1,1,0,0,0,0,0,0,0,0,0),
  CAREER_TECH = c(0,0,1,1,0,0,0,0,1,1,0,1,1),
  ENVIRONMENTAL = c(0,0,0,0,0,0,0,0,0,0,1,0,0)
)

magnet_programs$num_programs <- rowSums(magnet_programs[,2:7])
```

```r
# 4. MERGE PROGRAM DATA
full_data <- race_school_pct %>%
  left_join(magnet_programs, by = "SCHOOL_NAME") %>%
  mutate(
    STEM = coalesce(STEM, 0),
    ARTS = coalesce(ARTS, 0),
    IB = coalesce(IB, 0),
    MEDICAL = coalesce(MEDICAL, 0),
    CAREER_TECH = coalesce(CAREER_TECH, 0),
    ENVIRONMENTAL = coalesce(ENVIRONMENTAL, 0),
    num_programs = coalesce(num_programs, 0)
  )

full_data
```

```
## # A tibble: 606 x 18
##    SCHOOL_NAME    SCHOOL_YEAR race_df magnet traditional total_race total_school
##    <chr>                <dbl> <fct>    <dbl>       <dbl>      <dbl>        <dbl>
##  1 ALONSO HIGH S~        2020 White       46         668        714         2413
##  2 ALONSO HIGH S~        2020 Asian       16         100        116         2413
##  3 ALONSO HIGH S~        2020 Black        0         157        157         2413
##  4 ALONSO HIGH S~        2020 Hispan~     52        1273       1325         2413
##  5 ALONSO HIGH S~        2020 Indige~      0           0          0         2413
##  6 ALONSO HIGH S~        2020 Multir~      0         101        101         2413
##  7 ALONSO HIGH S~        2021 White       72         638        710         2516
##  8 ALONSO HIGH S~        2021 Asian       36          93        129         2516
##  9 ALONSO HIGH S~        2021 Black        0         159        159         2516
## 10 ALONSO HIGH S~        2021 Hispan~     92        1321       1413         2516
## # i 596 more rows
## # i 11 more variables: pct_race <dbl>, pct_magnet <dbl>, pct_traditional <dbl>,
## #   dominant_race <fct>, STEM <dbl>, ARTS <dbl>, IB <dbl>, MEDICAL <dbl>,
## #   CAREER_TECH <dbl>, ENVIRONMENTAL <dbl>, num_programs <dbl>
```

```r
# 5. Merge SES data
ses_school <- mt %>%
  filter(DEMOGRAPHIC_CATEGORY == "Meal Status") %>%

  group_by(SCHOOL_NAME, SCHOOL_YEAR, DEMOGRAPHIC) %>%
  summarise(
    total = sum(MAGNET_STUDENTS + TRADITIONAL_STUDENTS),
    .groups = "drop"
  ) %>%

  group_by(SCHOOL_NAME, SCHOOL_YEAR) %>%
  mutate(
    total_school = sum(total),
    pct_ses = total / total_school
  ) %>%
  ungroup() %>%

  mutate(
    ses_clean = case_when(
      grepl("free", DEMOGRAPHIC, ignore.case = TRUE) ~ "Free",
```

```r
      grepl("non", DEMOGRAPHIC, ignore.case = TRUE) ~ "Non-Free",
      TRUE ~ NA_character_
    )
  )

ses_school
```

```
## # A tibble: 109 x 7
##    SCHOOL_NAME      SCHOOL_YEAR DEMOGRAPHIC total total_school pct_ses ses_clean
##    <chr>                  <dbl> <chr>       <dbl>        <dbl>   <dbl> <chr>
##  1 ALONSO HIGH SCH~        2020 FREE/REDUC~  1515         2431   0.623 Free
##  2 ALONSO HIGH SCH~        2020 NON FREE/R~   916         2431   0.377 Free
##  3 ALONSO HIGH SCH~        2021 FREE/REDUC~  1338         2529   0.529 Free
##  4 ALONSO HIGH SCH~        2021 NON FREE/R~  1191         2529   0.471 Free
##  5 ALONSO HIGH SCH~        2022 FREE/REDUC~  2680         2680   1     Free
##  6 ALONSO HIGH SCH~        2023 FREE/REDUC~  2741         2741   1     Free
##  7 ARMWOOD HIGH SC~        2020 FREE/REDUC~  1716         2092   0.820 Free
##  8 ARMWOOD HIGH SC~        2020 NON FREE/R~   376         2092   0.180 Free
##  9 ARMWOOD HIGH SC~        2021 FREE/REDUC~  1633         2059   0.793 Free
## 10 ARMWOOD HIGH SC~        2021 NON FREE/R~   426         2059   0.207 Free
## # i 99 more rows
```

```r
ses_school <- ses_school %>%
  mutate(
    ses_clean = case_when(
      grepl("non", DEMOGRAPHIC, ignore.case = TRUE) ~ "Non-Free",
      grepl("free", DEMOGRAPHIC, ignore.case = TRUE) ~ "Free",
      TRUE ~ NA_character_
    )
  )

ses_wide <- ses_school %>%
  filter(!is.na(ses_clean)) %>%

  group_by(SCHOOL_NAME, SCHOOL_YEAR, ses_clean) %>%
  summarise(
    pct_ses = sum(pct_ses),
    .groups = "drop"
  ) %>%

  pivot_wider(
    names_from = ses_clean,
    values_from = pct_ses,
    values_fill = list(pct_ses = 0)
  )

ses_wide <- ses_wide %>%
  rename(
    pct_free = Free,
    pct_nonfree = `Non-Free`
  )

# 6. Final dataset
final_data <- full_data %>%
```

```r
  left_join(ses_wide, by = c("SCHOOL_NAME", "SCHOOL_YEAR"))



# 7. DIVERSITY INDEX
diversity_index <- race_school_pct %>%
  group_by(SCHOOL_NAME, SCHOOL_YEAR) %>%
  summarise(
    diversity = -sum(pct_race * log(pct_race + 1e-9)),
    .groups = "drop"
  )

final_data <- final_data %>%
  left_join(diversity_index, by = c("SCHOOL_NAME", "SCHOOL_YEAR"))

final_data
```

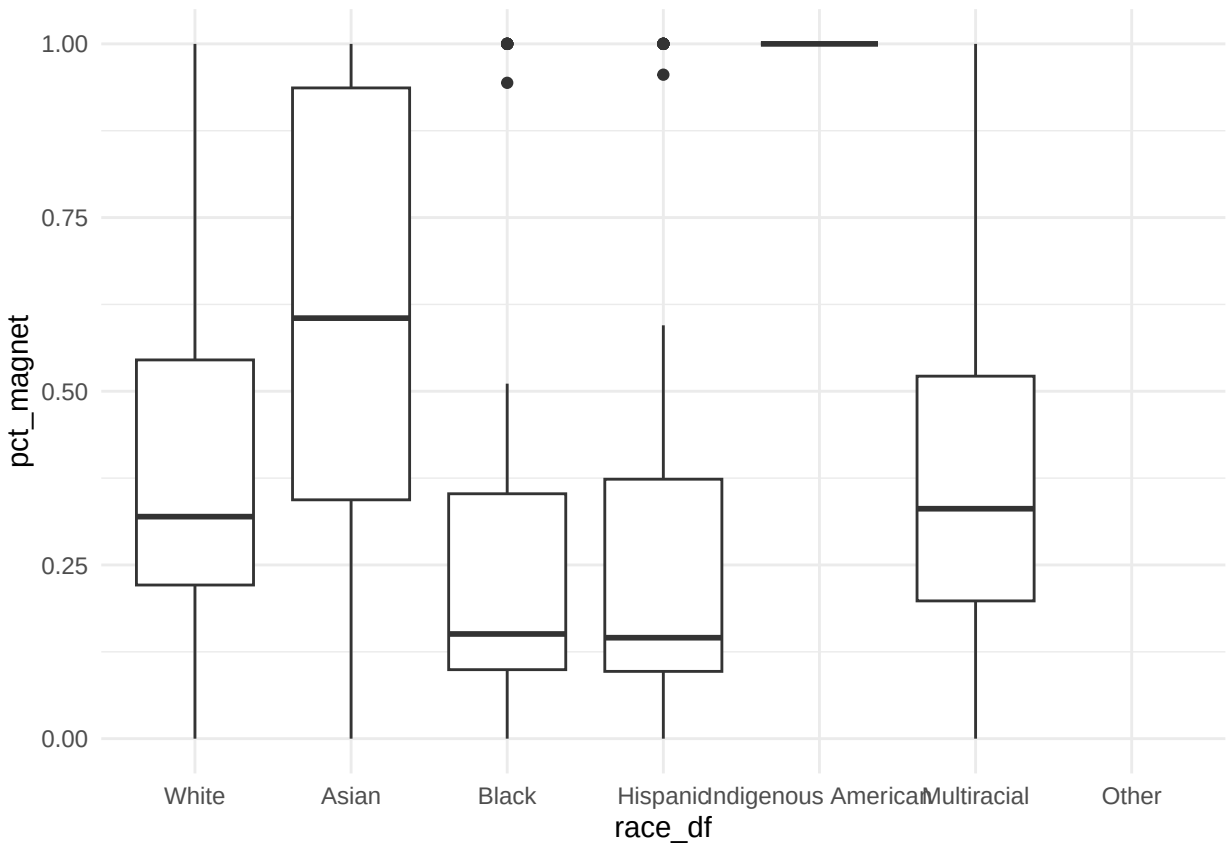
```
## # A tibble: 606 x 21
##    SCHOOL_NAME    SCHOOL_YEAR race_df magnet traditional total_race total_school
##    <chr>                <dbl> <fct>    <dbl>       <dbl>      <dbl>        <dbl>
##  1 ALONSO HIGH S~        2020 White       46         668        714         2413
##  2 ALONSO HIGH S~        2020 Asian       16         100        116         2413
##  3 ALONSO HIGH S~        2020 Black        0         157        157         2413
##  4 ALONSO HIGH S~        2020 Hispan~     52        1273       1325         2413
##  5 ALONSO HIGH S~        2020 Indige~      0           0          0         2413
##  6 ALONSO HIGH S~        2020 Multir~      0         101        101         2413
##  7 ALONSO HIGH S~        2021 White       72         638        710         2516
##  8 ALONSO HIGH S~        2021 Asian       36          93        129         2516
##  9 ALONSO HIGH S~        2021 Black        0         159        159         2516
## 10 ALONSO HIGH S~        2021 Hispan~     92        1321       1413         2516
## # i 596 more rows
## # i 14 more variables: pct_race <dbl>, pct_magnet <dbl>, pct_traditional <dbl>,
## #   dominant_race <fct>, STEM <dbl>, ARTS <dbl>, IB <dbl>, MEDICAL <dbl>,
## #   CAREER_TECH <dbl>, ENVIRONMENTAL <dbl>, num_programs <dbl>, pct_free <dbl>,
## #   pct_nonfree <dbl>, diversity <dbl>
```

```r
final_data_model <- final_data %>%
  filter(race_df != "Indigenous American")
final_data_model$race_df <- factor(final_data_model$race_df)

# 8. SCHOOL-LEVEL DATA (CRITICAL)
school_level <- final_data %>%
  group_by(SCHOOL_NAME, SCHOOL_YEAR) %>%
  summarise(
    pct_white = pct_race[race_df == "White"][1],
    num_programs = max(num_programs),
    pct_free = max(pct_free),
    diversity = max(diversity),
    .groups = "drop"
  ) %>%
  mutate(
    majority_white = ifelse(pct_white > 0.5, 1, 0)
  )
school_level
```

```
## # A tibble: 99 x 7
##    SCHOOL_NAME           SCHOOL_YEAR pct_white num_programs pct_free diversity
##    <chr>                       <dbl>     <dbl>        <dbl>    <dbl>     <dbl>
##  1 "ALONSO HIGH SCHOOL"         2020     0.296            1    0.623      1.15
##  2 "ALONSO HIGH SCHOOL"         2021     0.282            1    0.529      1.14
##  3 "ALONSO HIGH SCHOOL"         2022     0.271            1    1          1.14
##  4 "ALONSO HIGH SCHOOL"         2023     0.249            1    1          1.13
##  5 "ARMWOOD HIGH SCHOOL"        2020     0.247            0    0.820      1.32
##  6 "ARMWOOD HIGH SCHOOL"        2021     0.250            0    0.793      1.31
##  7 "ARMWOOD HIGH SCHOOL"        2022     0.237            0    1          1.30
##  8 "ARMWOOD HIGH SCHOOL"        2023     0.225            0    1          1.30
##  9 "Alonso High School"         2024     0.241            0   NA          1.10
## 10 "Alonso High School "        2025     0.234            0   NA          1.11
## # i 89 more rows
## # i 1 more variable: majority_white <dbl>
```

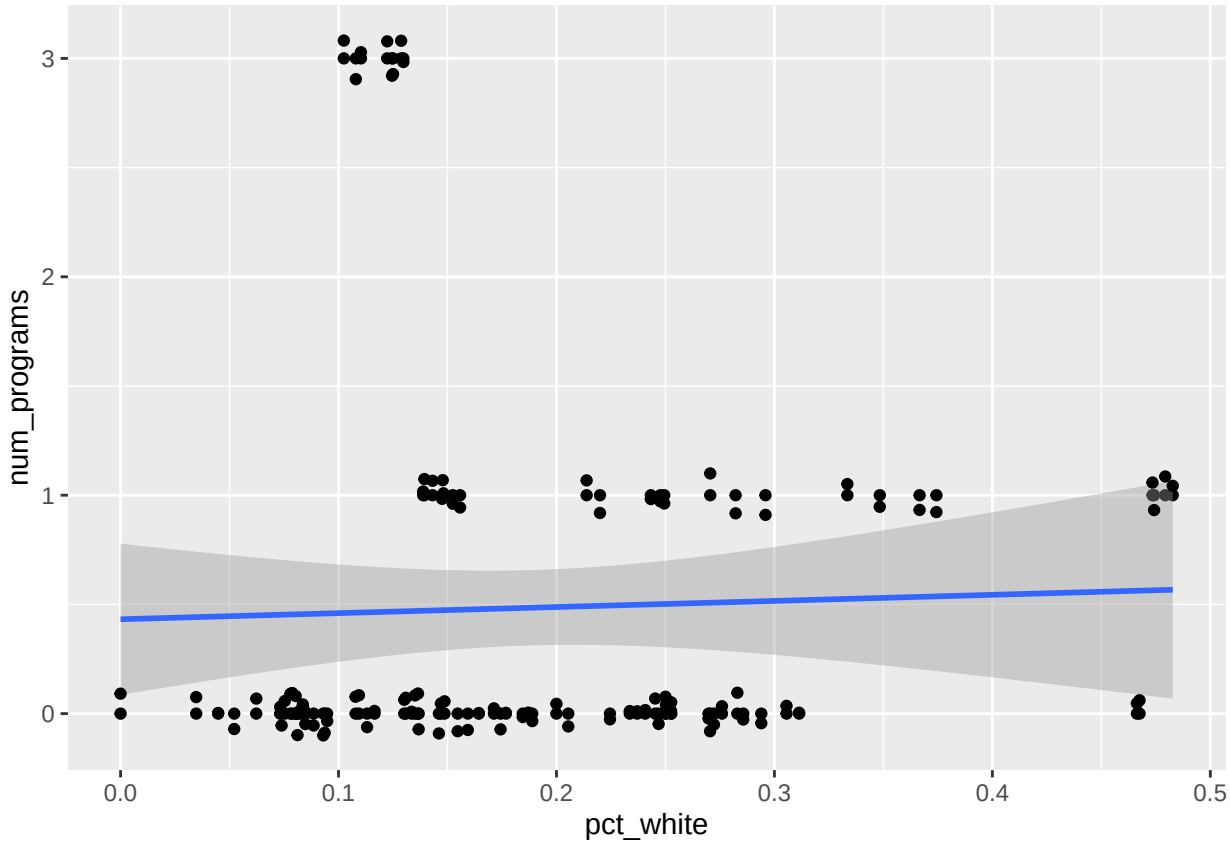
```r
# 9. VISUALS

# Race vs Magnet
ggplot(race_school_pct, aes(x = race_df, y = pct_magnet)) +
  geom_boxplot() +
  theme_minimal()
```

```
## Warning: Removed 134 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



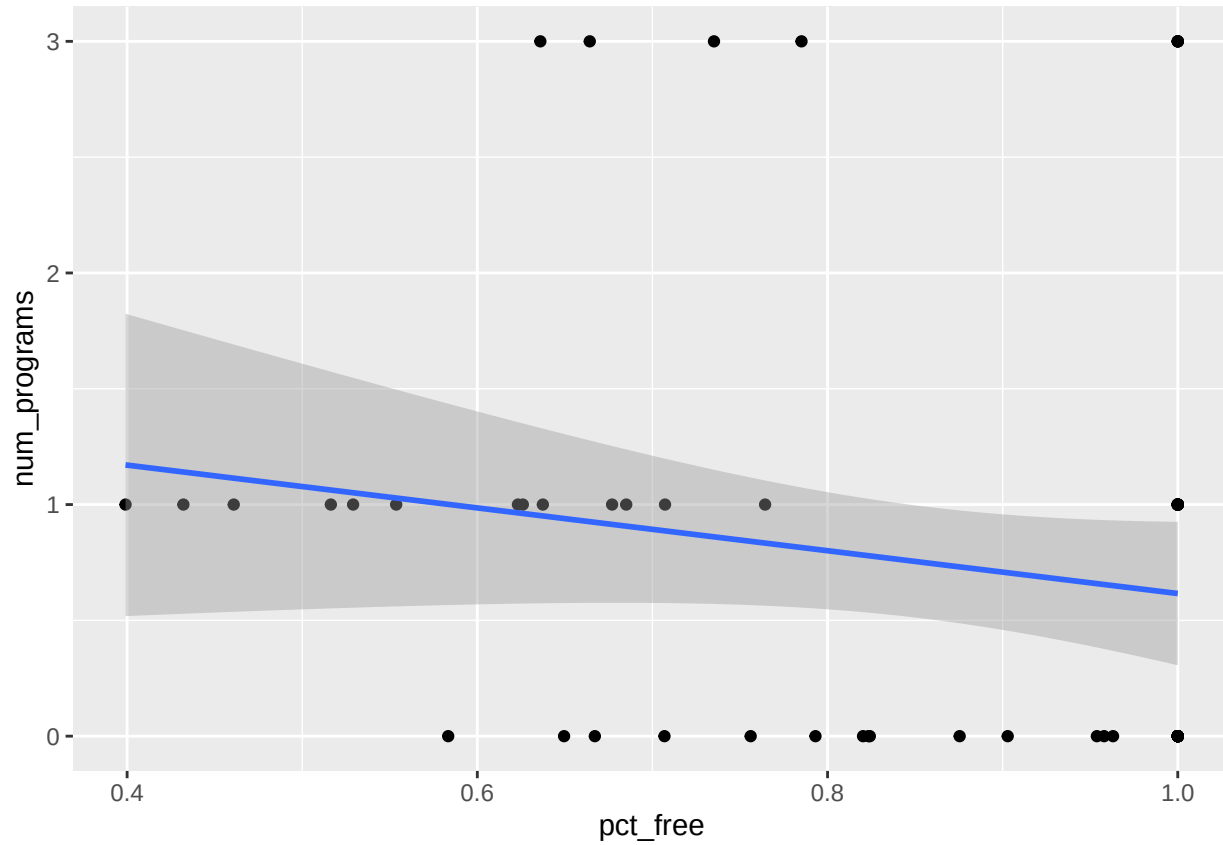
```r
# Program diversity vs race composition
ggplot(school_level, aes(x = pct_white, y = num_programs)) +
  geom_point() +
  geom_smooth(method = "lm")+
  geom_jitter(height = 0.1)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# SES vs magnet
ggplot(school_level, aes(x = pct_free, y = num_programs)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 35 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 35 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
# 10. MODELS
log.obj <- glm(
  cbind(magnet, traditional) ~ race_df,
  family = binomial,
  data = final_data_model
)
summary(log.obj)
```

```
##
## Call:
## glm(formula = cbind(magnet, traditional) ~ race_df, family = binomial,
##     data = final_data_model)
##
## Coefficients: (1 not defined because of singularities)
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)         -0.59047    0.01185 -49.842  < 2e-16 ***
## race_dfAsian         1.76067    0.02754  63.933  < 2e-16 ***
## race_dfBlack        -0.20862    0.01584 -13.168  < 2e-16 ***
## race_dfHispanic     -0.75475    0.01532 -49.255  < 2e-16 ***
## race_dfMultiracial   0.16480    0.02753   5.987 2.14e-09 ***
## race_dfOther              NA         NA      NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```
##     Null deviance: 64164  on 470  degrees of freedom
## Residual deviance: 52394  on 466  degrees of freedom
## AIC: 54507
##
## Number of Fisher Scoring iterations: 5
```

```r
log.obj.2 <- glm(
  cbind(magnet, traditional) ~ race_df + pct_free,
  family = binomial,
  data = final_data_model
)


log.obj.3 <- glm(
  cbind(magnet, traditional) ~ race_df + num_programs,
  family = binomial,
  data = final_data_model
)
summary(log.obj.3)
```

```
##
## Call:
## glm(formula = cbind(magnet, traditional) ~ race_df + num_programs,
##     family = binomial, data = final_data_model)
##
## Coefficients: (1 not defined because of singularities)
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)        -0.921240   0.012681 -72.648  < 2e-16 ***
## race_dfAsian        1.706836   0.028116  60.707  < 2e-16 ***
## race_dfBlack       -0.340489   0.016497 -20.640  < 2e-16 ***
## race_dfHispanic    -0.711639   0.015678 -45.391  < 2e-16 ***
## race_dfMultiracial  0.123947   0.028410   4.363 1.28e-05 ***
## race_dfOther              NA         NA      NA       NA
## num_programs        0.581267   0.006452  90.097  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 64164  on 470  degrees of freedom
## Residual deviance: 43941  on 465  degrees of freedom
## AIC: 46055
##
## Number of Fisher Scoring iterations: 5
```

```r
# head(final_data_model)


log.obj.int <- glm(
  cbind(magnet, traditional) ~ race_df + pct_free + num_programs + race_df:pct_free,
  family = binomial,
  data = final_data_model
)
summary(log.obj.int)
```

```
##
## Call:
## glm(formula = cbind(magnet, traditional) ~ race_df + pct_free +
##     num_programs + race_df:pct_free, family = binomial, data = final_data_model)
##
## Coefficients:
##                              Estimate Std. Error z value Pr(>|z|)
## (Intercept)                 -1.512825   0.057908 -26.125  < 2e-16 ***
## race_dfAsian                 1.914609   0.139429  13.732  < 2e-16 ***
## race_dfBlack                -1.682377   0.098457 -17.087  < 2e-16 ***
## race_dfHispanic             -1.214531   0.084782 -14.325  < 2e-16 ***
## race_dfMultiracial          -0.169326   0.145908  -1.161   0.2458
## pct_free                     0.334672   0.069499   4.815 1.47e-06 ***
## num_programs                 0.795556   0.007883 100.920  < 2e-16 ***
## race_dfAsian:pct_free       -0.420292   0.167786  -2.505   0.0122 *
## race_dfBlack:pct_free        1.282188   0.114308  11.217  < 2e-16 ***
## race_dfHispanic:pct_free     0.595168   0.099579   5.977 2.27e-09 ***
## race_dfMultiracial:pct_free  0.247652   0.174064   1.423   0.1548
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 42946  on 305  degrees of freedom
## Residual deviance: 23899  on 295  degrees of freedom
##   (203 observations deleted due to missingness)
## AIC: 25293
##
## Number of Fisher Scoring iterations: 5
```

```r
log.obj.full <- glm(
  cbind(magnet, traditional) ~ race_df + pct_free + num_programs + diversity,
  family = binomial,
  data = final_data_model
)
summary(log.obj.full)
```

```
##
## Call:
## glm(formula = cbind(magnet, traditional) ~ race_df + pct_free +
##     num_programs + diversity, family = binomial, data = final_data_model)
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)         -0.157881   0.079666  -1.982   0.0475 *
## race_dfAsian         1.616601   0.035206  45.918   <2e-16 ***
## race_dfBlack        -0.614163   0.021576 -28.465   <2e-16 ***
## race_dfHispanic     -0.824028   0.020308 -40.576   <2e-16 ***
## race_dfMultiracial   0.019825   0.035613   0.557   0.5777
## pct_free             0.552280   0.041960  13.162   <2e-16 ***
## num_programs         0.849948   0.008331 102.022   <2e-16 ***
## diversity           -1.259314   0.052191 -24.129   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 42946  on 305  degrees of freedom
## Residual deviance: 23496  on 298  degrees of freedom
##   (203 observations deleted due to missingness)
## AIC: 24884
##
## Number of Fisher Scoring iterations: 5
```

```r
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```r
vif(log.obj.full)
```

```
##                  GVIF Df GVIF^(1/(2*Df))
## race_df      1.162916  4        1.019045
## pct_free     1.088476  1        1.043300
## num_programs 1.187409  1        1.089683
## diversity    1.292157  1        1.136731
```

```r
cor(final_data_model$pct_free, final_data_model$pct_race, use = "complete.obs")
```

```
## [1] 2.808969e-18
```

```r
cor(final_data_model$pct_free, final_data_model$diversity, use = "complete.obs")
```

```
## [1] -0.3446834
```

```r
cor(final_data_model$pct_free, final_data_model$num_programs, use = "complete.obs")
```

```
## [1] -0.1736818
```

```r
cor(final_data_model$diversity, final_data_model$pct_race, use = "complete.obs")
```

```
## [1] 0.00728821
```

```r
cor(final_data_model$diversity, final_data_model$num_programs, use = "complete.obs")
```

```
## [1] 0.3360126
```

```r
# 11. ANOVA- Model COmparison
anova(log.obj.int, log.obj.full, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: cbind(magnet, traditional) ~ race_df + pct_free + num_programs +
##     race_df:pct_free
## Model 2: cbind(magnet, traditional) ~ race_df + pct_free + num_programs +
##     diversity
```

```
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1       295      23899
## 2       298      23496 -3   402.73
```

Validation

```r
# -------------------------------
# STEP 1: Clean + fix factors FIRST
# -------------------------------

final_data_model <- final_data_model %>%
  filter(race_df != "Other")   # remove empty category

final_data_model$race_df <- factor(final_data_model$race_df)

# -------------------------------
# STEP 2: Train/Test Split
# -------------------------------

set.seed(123)

train_idx <- sample(1:nrow(final_data_model), size = 0.7*nrow(final_data_model))

train <- final_data_model[train_idx, ]
test  <- final_data_model[-train_idx, ]

# Ensure SAME factor levels
test$race_df <- factor(test$race_df, levels = levels(train$race_df))

# -------------------------------
# STEP 3: Fit Model (TRAIN ONLY)
# -------------------------------

log.obj.full <- glm(
  cbind(magnet, traditional) ~ race_df + pct_free + num_programs + diversity,
  family = binomial,
  data = train
)

# -------------------------------
# STEP 4: Predict (TEST ONLY)
# -------------------------------

test$pred <- predict(log.obj.full, newdata = test, type = "response")

# -------------------------------
# STEP 5: Create Actual Values
# -------------------------------

test$total <- test$magnet + test$traditional
test$actual <- test$magnet / test$total

test <- test %>%
```

```r
  filter(
    !is.na(pred),
    !is.na(actual),
    total > 0
  )

# -------------------------------
# STEP 6: Metrics
# -------------------------------

# MAE
mae <- mean(abs(test$pred - test$actual))

# RMSE
rmse <- sqrt(mean((test$pred - test$actual)^2))

# Weighted MAE (IMPORTANT)
weighted_mae <- sum(test$total * abs(test$pred - test$actual)) / sum(test$total)

eps <- 1e-15

log_loss <- -mean(
  test$actual * log(test$pred + eps) +
  (1 - test$actual) * log(1 - test$pred + eps)
)

log_loss
```

```
## [1] 0.6090082
```

```r
mae
```

```
## [1] 0.2213672
```

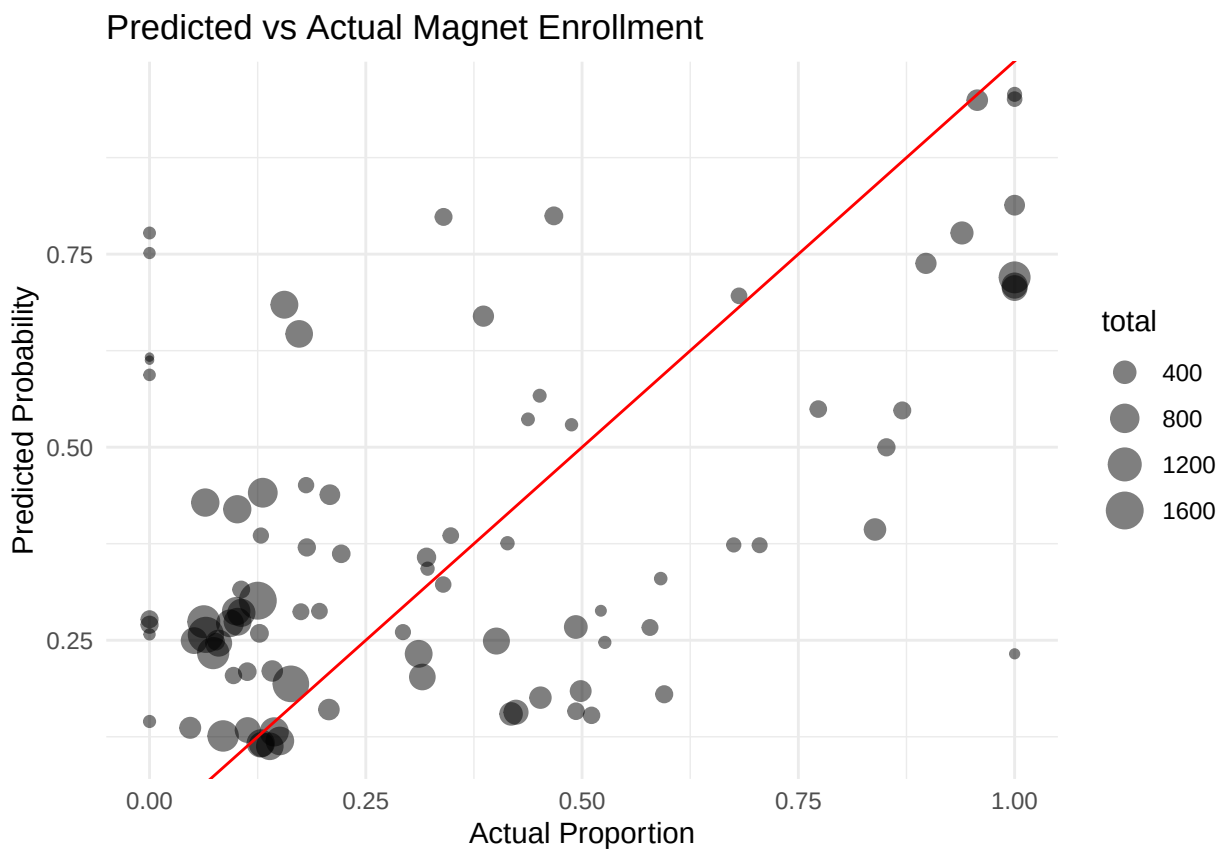
```r
rmse
```

```
## [1] 0.2820449
```

```r
weighted_mae
```

```
## [1] 0.1868362
```

```r
ggplot(test, aes(x = actual, y = pred, size = total)) +
  geom_point(alpha = 0.5) +
  geom_abline(slope = 1, intercept = 0, color = "red") +
  labs(
    title = "Predicted vs Actual Magnet Enrollment",
    x = "Actual Proportion",
    y = "Predicted Probability"
  ) +
  theme_minimal()
```




```r
write.csv(final_data_model, "MagnetTraditional.csv", row.names = FALSE)
```

Final Conclusion

The objective of this analysis was to determine whether disparities in magnet school enrollment are driven by race, socioeconomic status (SES), or underlying structural differences across schools. To address this, a sequence of models was constructed, beginning with a race-only specification and progressively incorporating SES, program availability, diversity, and interaction effects.

The race-only model established that enrollment probabilities differ significantly across racial groups. However, this model is inherently incomplete. By treating race as the sole predictor, it implicitly attributes all observed differences to race itself, without accounting for variation in school-level opportunity. This creates a classic underfitting problem, where omitted structural variables bias interpretation and obscure the true mechanism driving disparities.

To correct for this, additional variables were introduced. The full structural model (`log.obj.full`) incorporated SES (measured as the proportion of students receiving free lunch), the number of available magnet programs, and a diversity index capturing racial composition within schools. The results from this model demonstrate that structural factors dominate the enrollment process.

Program Availability Effect

The coefficient for program availability is:

$$\hat{\beta}_{programs} = 0.849948$$

$$e^{0.849948} = 2.339$$

Thus, each additional magnet program increases the odds of enrollment by:

$$2.339 - 1 = 1.339$$

or **133.9% per additional program**.

This is the largest effect observed across all predictors, indicating that access to magnet programs is primarily determined by institutional availability rather than individual characteristics. In practical terms, students in schools offering more programs are dramatically more likely to enroll, regardless of race.

SES Effect

The SES effect is also statistically significant:

$$\hat{\beta}_{pct_free} = 0.552280$$

$$e^{0.552280} = 1.737$$

This corresponds to a **73.7% increase in odds for a full-unit increase in SES**. However, because SES is measured on a 0–1 scale, a more meaningful interpretation is:

$$e^{0.552280 \cdot 0.1} \approx 1.057$$

or approximately a **5.7% increase in odds for a 10 percentage point increase in poverty concentration**.

This indicates that SES plays a role, but its magnitude is substantially smaller than the effect of program availability.

Diversity Effect

The diversity coefficient is:

$$\hat{\beta}_{diversity} = -1.259314$$

$$e^{-1.259314} = 0.284$$

$$1 - 0.284 = 0.716$$

An increase in school-level diversity is associated with a **71.6% decrease in the odds of magnet enrollment**.

This suggests that more racially heterogeneous schools tend to have lower levels of magnet participation, reinforcing the importance of structural differences across schools.

Interaction Model Interpretation

While the structural model provides a strong baseline, the interaction model (`log.obj.int`) reveals that the relationship between SES and enrollment is not uniform across racial groups.

For White students (reference group):

$$\hat{\beta}_{SES} = 0.334672$$

$$e^{0.334672} = 1.398$$

This corresponds to a **39.8% increase in odds**.

Black Students

Interaction term:

$$\hat{\beta}_{Black \times SES} = 1.282188$$

Total SES effect:

$$0.334672 + 1.282188 = 1.61686$$

$$e^{1.61686} = 5.04$$

This corresponds to a **five-fold increase in odds**.

This indicates that SES improvements translate into substantially larger gains in access for Black students relative to White students.

Hispanic Students

$$\hat{\beta}_{Hispanic \times SES} = 0.595168$$

$$e^{0.595168} = 1.813$$

This implies an **81.3% stronger SES effect relative to White students**.

These results demonstrate that SES does not operate uniformly across racial groups. Instead, it acts as an amplifying mechanism for some groups and a weaker mechanism for others.

Model Comparison (ANOVA)

The analysis of deviance between the interaction model and the structural model yields:

$$\chi^2 = 402.73$$

This large improvement in fit indicates that interaction effects are essential for accurately modeling enrollment probabilities. Ignoring these interactions results in model misspecification.

Core Findings

Taken together, the results establish three key findings:

1. Race remains a statistically significant predictor even after controlling for SES and structural variables.
2. Structural factors—particularly the number of available magnet programs—are the dominant drivers of enrollment.
3. SES operates differently across racial groups, meaning gains in socioeconomic status do not translate equally across groups.

Final Model Validation Conclusion IMPORTANT NOTE

The model demonstrates moderate predictive performance, with an average error of approximately 22 percentage points and improved accuracy for larger schools. However, systematic deviations in the predicted vs. actual plot reveal that the model consistently underpredicts high-enrollment schools and overpredicts low-enrollment schools. This compression toward the mean indicates that important determinants of magnet enrollment remain unobserved. While structural variables such as program availability significantly improve model fit, the remaining prediction error suggests that institutional constraints—such as program capacity, admissions criteria, and geographic access—play a critical role in shaping enrollment outcomes.

Final Interpretation

The results can be summarized as:

$$\text{Observed inequality} = \text{Structural access differences} + \text{Differential SES effects}$$

not simply differences in individual behavior.

The primary driver of disparities in magnet school enrollment is not race alone, nor SES alone, but the interaction between demographic characteristics and unequal access to educational opportunities across schools. Therefore, Magnet enrollment is fundamentally structured by access. Students are not making choices within a uniform system; they are operating within different opportunity environments. These environments—defined by program availability, school composition, and socioeconomic context—ultimately determine who has access to magnet programs and who does not.